$$\frac{dx(t)}{dt} = v' = \omega y(t) \qquad \frac{d^2x(t)}{dt^2} = -\omega^2 x(t)$$

$$\frac{dy(t)}{dt} = v^2 = -\omega x(t)$$

$$x''(t) + \omega^2 x(t) = 0.$$